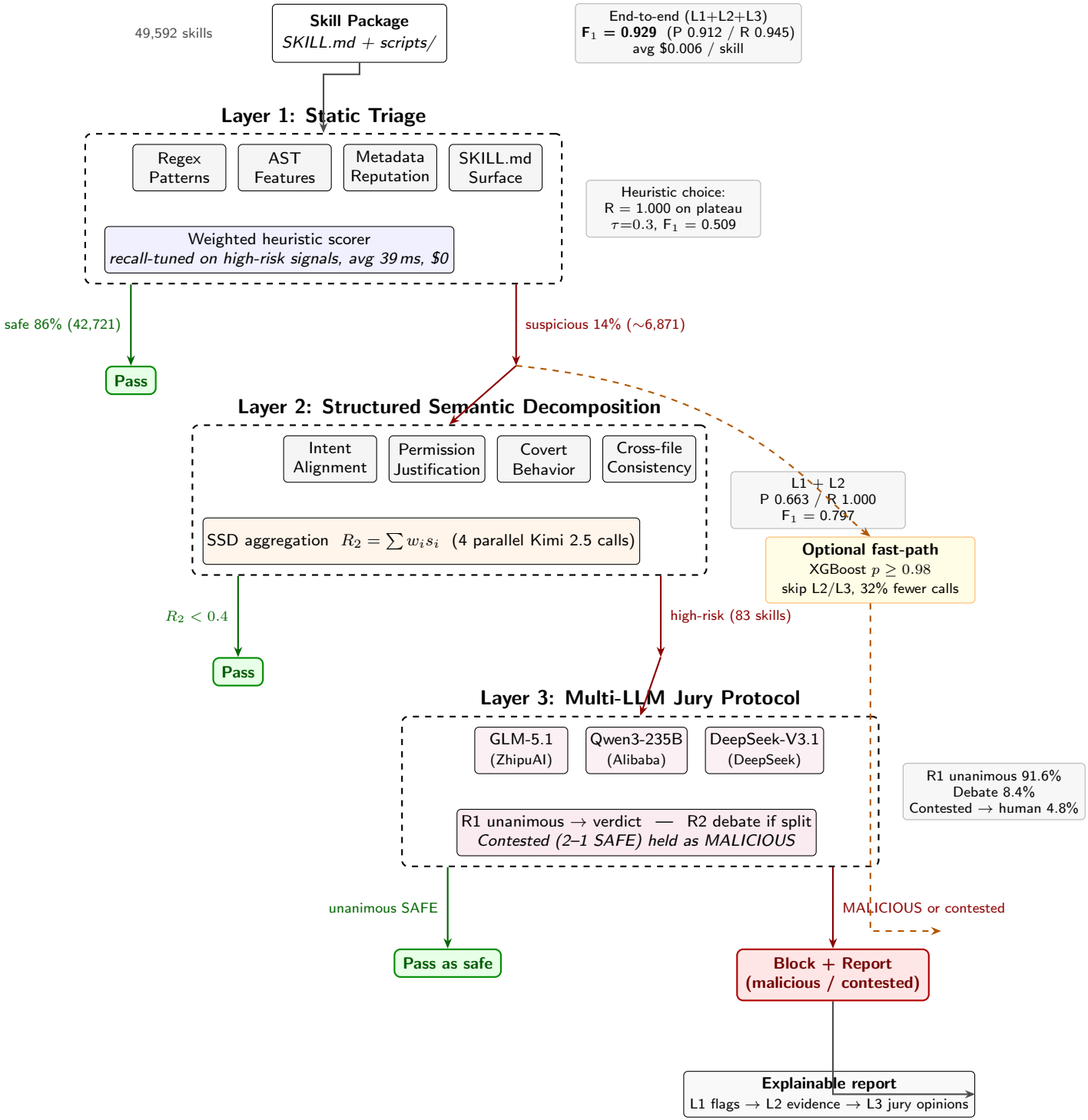

49,592 skills
Skill Package
SKILL.md + scripts/
End-to-end (L1+L2+L3)
F1 = 0.929 (P 0.912 / R 0.945)
avg $0.006 / skill
Layer 1: Static Triage
Regex Patterns
AST Features
Metadata Reputation
SKILL.md Surface
Weighted heuristic scorer
recall-tuned on high-risk signals, avg 39 ms, $0
Heuristic choice:
R = 1.000 on plateau
τ=0.3, F1 = 0.509
safe 86% (42,721)
suspicious 14% (~6,871)
Pass
Layer 2: Structured Semantic Decomposition
Intent Alignment
Permission Justification
Covert Behavior
Cross-file Consistency
SSD aggregation $R_2 = \sum w_i s_i$ (4 parallel Kimi 2.5 calls)
L1 + L2
P 0.663 / R 1.000
F1 = 0.797
Optional fast-path
XGBoost $p \geq 0.98$
skip L2/L3, 32% fewer calls
$R_2 < 0.4$
high-risk (83 skills)
Pass
Layer 3: Multi-LLM Jury Protocol
GLM-5.1 (ZhipuAI)
Qwen3-235B (Alibaba)
DeepSeek-V3.1 (DeepSeek)
R1 unanimous 91.6%
Debate 8.4%
Contested → human 4.8%
R1 unanimous → verdict — R2 debate if split
Contested (2–1 SAFE) held as MALICIOUS
unanimous SAFE
MALICIOUS or contested
Pass as safe
Block + Report (malicious / contested)
Explainable report
L1 flags → L2 evidence → L3 jury opinions